SOLITEC



POWER THAT STAYS ON

ENERGY STORAGE SOLUTIONS

Hybrid inverters & LFP batteries · **Residential** · **Commercial**

Cross-scenario energy solutions

One integrated family of hybrid inverters and lithium iron phosphate batteries, from a 3 kW home system to 180 kW commercial installations — supplied, installed and supported by Olitec across Tamil Nadu and South India.

Hybrid Inverter Series



OLI-1H · Single-phase 3 – 6 kW

Low-voltage (48 V) battery · 2 MPPT · IP66 · fan-less

OLI-1H3L · 3.6L · 4L · 4.6L · 5L · 6L · 5L-2

200 % PV oversizing | <10 ms backup



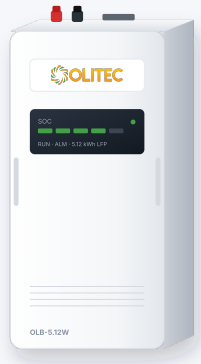
OLI-3H · Three-phase 15 – 30 kW

High-voltage (160 – 750 V) battery · up to 3 MPPT · IP66

OLI-3H15H · 20H · 25H · 29.9H · 30H

60 kW PV input | 6 units parallel

LFP Battery Series

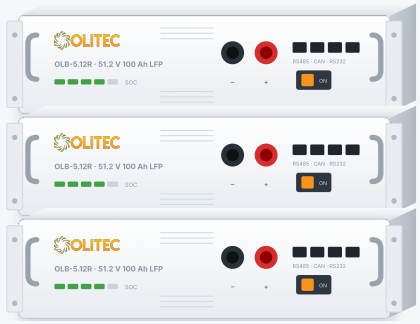


OLB-5.12W · Wall-mount 5.12 kWh

51.2 V · 100 Ah · 644 × 410 × 198 mm · 54 kg

Slim wall pack for homes and small offices

4,000 cycles | Up to 10 in parallel



OLB-5.12R · Rack-mount 5.12 kWh

51.2 V · 100 Ah · 3U / 19″ · 45 kg

Stack in a standard cabinet: 5.12 – 51.2 kWh

LiFePO$_4$ · BIS cells | RS485 / CAN

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HYBRID INVERTER SERIES

Tailor-made power flexibility

Flexible compatibility across solar, battery and grid

Up to 200 % PV input power
Higher PV input tolerance on every MPPT

Up to 200 % grid input power
Flexible grid charging for heavy loads

DC input current up to 18 A
Easier pairing with high-current modules

Up to 6 units in parallel
Scales as your needs grow

Universal battery compatibility
Olitec OLB packs plug-and-play; third-party lithium via CAN

10 ms seamless transition
Loads never notice an outage

INTELLIGENT POWER CAPTURE & MANAGEMENT

VPP access
Virtual power plant ready

Adaptive control
Dynamic tariff response

6-period TOU
Time-of-use scheduling

Peak shaving
Grid-constraint aware

Shadow scan
Optimal production

8-level protection
AFCI · SPD · leakage

LFP BATTERY SERIES

All-around safety

Total protection from every angle

$LiFePO_4$

- Lithium iron phosphate cells with superior thermal stability — no cobalt, non-toxic, non-polluting.
- Cell-level certification: BIS, UN 38.3, IEC 62619, UL 1973.
- Built-in BMS with over-charge, over-discharge, over-current, short-circuit and temperature protection.
- Passive cooling — no fans, nothing to service.

LFP BATTERY SERIES

Instant scalability, enduring power

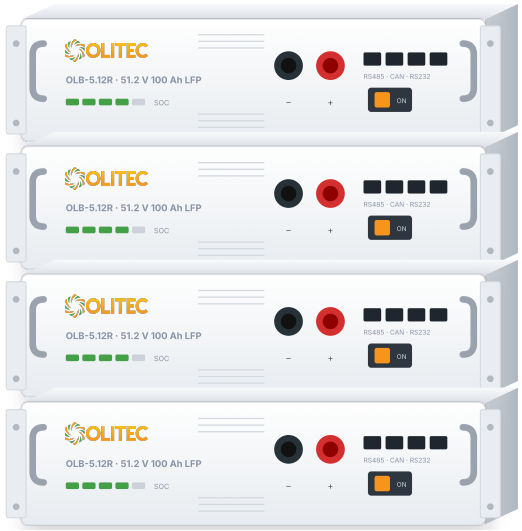
Modular design, grow with your needs

51.2 kWh

Up to ten OLB-5.12 packs in parallel on one 51.2 V bus — from 5.12 kWh for a small home to 51.2 kWh for a clinic or office.

4,000 cycles

Rated at 80 % depth of discharge, 25 °C, 0.5C — over a decade of daily cycling.



OLB-5.12R · four packs = 20.48 kWh in 12U



Simplicity, aesthetic & hassle-free installation

The 198 mm-slim wall-mount OLB-5.12W fits beside the inverter in a utility corner; the rack-mount OLB-5.12R drops into any standard 19″ cabinet with tool-free paralleling via the module combiner box.

Dependable energy management

The BMS reports state of charge, health and alarms to the Olitec OLI-1H inverter over CAN, with RS485 and RS232 for third-party systems and monitoring — safer, longer-lasting energy on tap.

Built for every scenario

Grid-interactive storage that pays for itself twice: once through the outages it covers, and again through the tariff it avoids.

Residential backup & savings

OLI-1H 3 – 6 kW · OLB-5.12 × 1 – 4

- Whole-home or essential-load backup with < 10 ms switch-over.
- Self-consume rooftop solar in the evening instead of exporting at low feed-in rates.
- Works with PM Surya Ghar rooftop systems; expandable as loads grow.

Small commercial & institutions

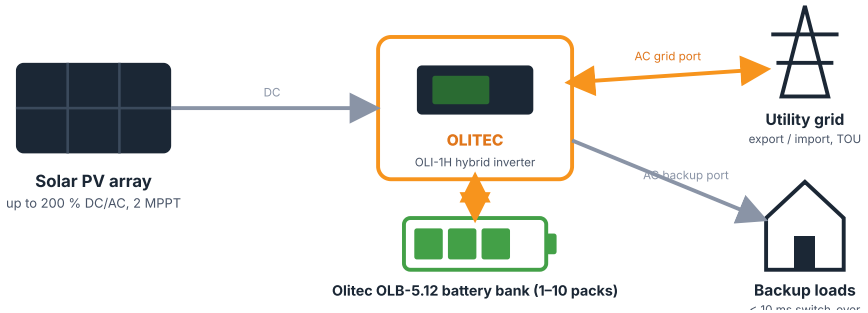
OLI-1H × up to 6 parallel · OLB-5.12 × up to 10

- Clinics, schools, shops and offices on single-phase supply.
- Time-of-use scheduling shifts consumption out of peak-tariff windows.
- Replace diesel generator run-hours with silent, zero-emission storage.

Commercial & industrial

OLI-3H 15 – 30 kW · high-voltage battery

- Three-phase hybrid with up to 60 kW PV and dual battery inputs.
- Peak shaving to cap contracted demand; 100 % unbalanced-phase output.
- AC coupling to existing PV, generator integration and VPP participation.



How it works

- **Day:** PV powers the loads; surplus charges the battery, then exports to the grid.
- **Evening peak:** the battery supplies the home or business at the highest tariff hours.
- **Outage:** the backup port keeps essential circuits live while PV continues to charge.
- **Off-peak:** optional grid charging at the lowest tariff, scheduled from the app.

Single-Phase Hybrid Inverter 3 – 6 kW

OLI-1H · Low-voltage battery

Model	OLI-1H3L	OLI-1H3.6L	OLI-1H4L	OLI-1H4.6L	OLI-1H5L	OLI-1H6L	OLI-1H5L-2
DC Input (PV Side)							
Max. input power [W]	4800	5760	6400	7360	8000	9600	8000
Max. input voltage [V]	500						
MPPT voltage range [V]	125 – 425						
Max. input current per MPPT [A]	18.0 / 18.0						
No. of MPPTs / strings per MPPT	2 / 1+1						
Battery Connection							
Battery type	Lithium-ion / Lead-acid						
Max. charge / discharge power [W]	3000 / 3000	3600 / 3600	4000 / 4000	4600 / 4600	5000 / 5000	6000 / 6000	5000 / 5000
Battery voltage range [V]	40 – 60						
Max. charge / discharge current [A]	70	90	100	110	120	135	120
AC Output (Grid Port)							
Rated output power [W]	3000	3600	4000	4600	5000	6000	5000
Max. output apparent power [VA]	3300	4000	4400	5000	5500	6600	5000
Rated grid voltage / frequency	220 / 230 V, 50 / 60 Hz						
Max. output current [A]	15.0	16.0	20.0	22.7	25.0	30.0	22.7
AC Output (Backup Port)							
Max. output apparent power [VA]	3300	4000	4400	5000	5500	6600	5000
Max. output current [A]	15.0	18.2	20.0	22.7	25.0	30.0	22.7
Max. output peak power [W] / duration	6000 (<10 s)	7200 (<10 s)	8000 (<10 s)	9200 (<10 s)	10000 (<10 s)	10000 (<10 s)	10000 (<10 s)
Grid-tied to stand-alone switching time [ms]	< 10						
Efficiency							
Max. efficiency [%]	97.0	97.0	97.0	97.3	97.3	97.3	97.3
Euro efficiency [%]	96.0	96.0	96.0	96.2	96.2	96.2	96.2
General Data							
Dimensions (W × H × D) [mm]	228 × 470 × 340						
Weight [kg]	19						
Operating temperature range [°C]	–20 to +60 (derating above +45)						
Ingress protection	IP66						
Cooling	Natural convection						
Standards & Warranty							
Standard warranty [years]	5 (extended warranty available)						

Condensed data — refer to the OLI-1H datasheet for the complete specification, protection functions and standards compliance. OLI-1H5L-2: 5 kW variant without 10 % AC-side overload capacity.

Three-Phase Hybrid Inverter 15 – 30 kW

OLI-3H · High-voltage battery

Model	OLI-3H15H	OLI-3H20H	OLI-3H25H	OLI-3H29.9H	OLI-3H30H
DC Input (PV Side)					
Max. input power [W]	30000	40000	50000	59800	60000
Max. input voltage [V]			1000		
MPPT voltage range [V]			160 – 850		
Max. input current per MPPT [A]	36 / 36	36 / 36	36 / 36 / 36	36 / 36 / 36	36 / 36 / 36
No. of MPPTs / strings per MPPT	2 / 2+2	2 / 2+2	3 / 2+2+2	3 / 2+2+2	3 / 2+2+2
Battery Connection					
Battery type			Lithium-ion (high-voltage)		
Max. charge / discharge power [W]	16500	22000	27500	30000	33000
Battery voltage range [V]			160 – 750		
Max. charge / discharge current [A]	50	50	50 + 50	50 + 50	50 + 50
Number of battery inputs	1	1	2	2	2
AC Output (Grid Port)					
Rated output power [W]	15000	20000	25000	29900	30000
Max. output apparent power [VA]	16500	22000	27500	29900	33000
Rated grid voltage / frequency			3L / N / PE, 220 / 380, 230 / 400 V, 50 / 60 Hz		
Max. output current [A]	25 / 24	33.4 / 31.9	41.7 / 39.9	45.4 / 43.4	50 / 47.9
AC Output (Backup Port)					
Max. output apparent power [VA]	16500	22000	27500	29900	33000
Max. output current [A]	25 / 24	33.4 / 31.9	41.7 / 39.9	45.4 / 43.4	50 / 47.9
Max. output peak power [W] / duration	30000 (<10 s)	40000 (<10 s)	40000 (<10 s)	48000 (<10 s)	48000 (<10 s)
Grid-tied to stand-alone switching time [ms]			< 10		
Efficiency					
Max. efficiency [%]			97.8		
Euro efficiency [%]	96.7	97.2	97.2	97.2	97.2
General Data					
Dimensions (W × H × D) [mm]			550 × 800 × 230		
Weight [kg]	46.5	46.5	50	50	50
Operating temperature range [°C]			−25 to +60		
Ingress protection			IP66		
Cooling			Intelligent fan cooling		
Standards & Warranty					
Standard warranty [years]			5 (extended warranty available)		

Condensed data — refer to the OLI-3H datasheet for the complete specification, protection functions and standards compliance.

LFP Battery 5.12 kWh · OLB-5.12 Series

51.2 V · 100 Ah

Pack	OLB-5.12W (Wall)	OLB-5.12R (Rack)
Nominal voltage	51.2 V	51.2 V
Nominal capacity	100 Ah	100 Ah
Rated energy	5.12 kWh	5.12 kWh
Operating voltage range	44.8 – 57.6 V	44.8 – 57.6 V
Recommended charge / discharge current	50 A continuous	50 A continuous
Max. charge / discharge current	100 A	100 A
Charging temperature range	0 °C to 50 °C	0 °C to 50 °C
Discharging temperature range	0 °C to 55 °C	0 °C to 55 °C
Cooling	Passive	Passive
Internal resistance	≤ 0.25 mΩ	≤ 0.25 mΩ
Cycle life	4,000 cycles @ 25 °C, 80 % DoD, 0.5C / 0.5C, to 70 % SoH	4,000 cycles @ 25 °C, 80 % DoD, 0.5C / 0.5C, to 70 % SoH
Dimensions (H × W × D)	644 × 410 × 198 mm	133.5 × 482 × 553 mm (3U, 19″ rack)
Weight	54 ± 2 kg	45 ± 2 kg
Ingress protection	IP20	IP20
Communication	RS485 × 3, CAN × 1, RS232 × 1	RS485 × 3, CAN × 1, RS232 × 1
Storage condition	15 – 35 °C; self-discharge ≈ 7 % / month; recharge every 3 months	15 – 35 °C; self-discharge ≈ 7 % / month; recharge every 3 months
Mounting	Wall-mounted	Rack-mounted

Cell	Specification
Cell chemistry	Lithium iron phosphate (LFP / $LiFePO_4$)
Nominal rating	3.2 V, 100 Ah (320 Wh per cell)
AC internal impedance	≤ 0.6 mΩ (0.3 mΩ typical)
Standard continuous charge / discharge current	50 A / 50 A (0.5C / 0.5C)
Max. continuous charge / discharge current	100 A / 100 A (1C / 1C)
Peak charge / discharge current (10 s, non-cyclic)	200 A / 300 A (2C / 3C)
Cell-level certification	BIS, UN 38.3, IEC 62619, UL 1973

Parallel system (up to 10 packs)	Specification
Nominal capacity	100 Ah × N (N = number of packs, max. 10)
Usable energy range	5.12 kWh to 51.2 kWh
Nominal voltage	51.2 V
Recommended charge / discharge current	50 A continuous
Max. charge / discharge current	100 A
Operating voltage	43.2 – 57.6 V
Operating temperature	0 – 50 °C (charging), 0 – 55 °C (discharging)
System weight	Number of packs × unit weight
Dimensions	As per general arrangement drawing / mounting layout
Parallel connection	Module power combiner box recommended for parallel systems

Cycle life: 4,000 cycles at 25 °C, 80 % DoD, 0.5C / 0.5C to 70 % SoH. A module power combiner box is recommended for parallel systems.

ABOUT OLITEC

Partner for all your solar energy needs

Olitec Solar Private Limited is the clean-energy business of the GSONS Group — the family behind the Butterfly Gandhimathi Appliances brand. Since 2016 we have delivered rooftop and open-access solar plants, manufactured PV modules and balance-of-system components, and supplied distributed-energy and storage products through our channel partners across South India.

- **Own-brand products:** Olitec hybrid inverters and LFP batteries, backed by local stock, service and warranty support from Chennai.
- **Turnkey EPC:** rooftop, ground-mount and open-access solar from 3 kW to utility scale, with O&M and energy audits.
- **Channel programme:** dealer and installer onboarding, training, design support and co-branded marketing.

2016
Clean-energy journey began; PV module manufacturing since 2017

500+
Rooftop customers served within two years of launch

3.7 crore kg
CO_2 emissions avoided by Olitec-built plants

₹39 crore
Revenue generated for our customers from solar

15 MW
Solar park commissioned; 100 MW park in development

1.5 lakh
Trees-equivalent planted

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Let's design your energy storage system.

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